A-CAT Corp: AI-Augmented MRP Planning Audit

Purpose of the exercise
This submission uses AI as a planning analyst rather than a final decision-maker. Part 1 is checking whether an AI-generated MRP matches my solution for A-CAT Corp's P0110 product family and documents how the prompt had to be refined to get there.
Part 2 goes further and questions whether the lot sizes in Exhibit 1 should be treated as fixed inputs or reviewed as a planning policy using the cost data in Exhibit 5.

Part 1: AI As-Is MRP Check

Summary for Manual MRP

The full 8-week, 6-row matrix for all 5 summarised is shown below. The summary below is the reference point for the deviation check.

Item	Lot size	Lead time	On-hand	Orders placed (8 weeks)	Ending inventory (Week 8)
P0110	613	2 weeks	240	5	305
P0X	1,226	1 week	500	4	500
P0Y	1,839	1 week	340	3	953
P1X1	2,452	1 week	500	3	500
P1X2	1,839	1 week	600	3	600

Prompt 1 (initial) - *Using the attached A-CAT Corp case exhibits (Bill of Material, Product Structure Diagram, Master Schedule, On-Hand Inventory, and Costs), prepare an 8-week material requirements planning (MRP) matrix for P0110, P0X, P0Y, P1X1, and P1X2. Use the fixed lot sizes given in Exhibit 1. Show gross requirements, scheduled receipts, projected available balance, net requirements, planned order receipts, and planned order releases for each item*

AI response 1 (summary)
The first response correctly exploded the bill of materials and multiplied requirements down the product tree, but it made three consequential errors: (1) for P0110, it offset the planned order release by one week instead of the two-week lead time given in Exhibit 1, so every release landed a week later than it should; (2) for every item, it recorded the Week 1 shortfall as a Planned Order Release in Week 1 itself rather than as a Scheduled Receipt, missing the exercise's Week 1 assumption; and (3) for P1X2, it ordered the exact net requirement (3,078 units) instead of rounding up to the fixed lot size of 1,839 units (which should call for two lots, 3,678 units).

Deviation table

Item / Row	Manual answer	AI answer	Deviation observed	Likely reason
P0110 - release timing for the Week 3 receipt	Week 1	Week 2	Release shown one week late	AI applied a 1-week lead time to every item instead of P0110's own 2-week lead time
All items - Week 1 order type	Scheduled receipt	Planned order release	Same quantity, wrong row	AI did not register the "assume Week 1 scheduled receipt" instruction as an override on normal MRP logic
P1X2 - planned order receipt, Week 3	3,678 (2 lots)	3,078	Ordered exact requirement, not a full lot	AI defaulted to lot-for-lot sizing and overlooked the fixed lot size of 1,839
P0Y - ending inventory, Week 4	340 (no order needed)	~1,953 (extra order placed)	Unnecessary order triggered	Earlier timing errors threw off the running balance by the time AI reached Week 4
P1X1 - number of orders across 8 weeks	3	4	One extra order	Cascading effect of the lead-time and Week 1 errors on P0X's releases, which drive P1X1's gross requirements

Prompt 2 - *Redo the MRP matrix with these corrections. First, use each item's own lead time exactly as stated in Exhibit 1 (2 weeks for P0110, 1 week for P0X, P0Y, P1X1, and P1X2) when calculating planned order releases: release week equals receipt week minus lead time. Second, for every item, if a net requirement exists in Week 1, show it as a Scheduled Receipt of one full lot arriving in Week 1, not as a Planned Order Release, since any release that would fall before Week 1 is already committed. Third, always round Planned Order Receipts up to the nearest multiple of the item's fixed lot size in Exhibit 1, never to the exact net requirement. Fourth, recompute gross requirements level by level (P0110, then P0X and P0Y, then P1X1 and P1X2), where a component's gross requirement equals its parent's planned order release multiplied by the quantity-per-parent in Exhibit 1. Show the full 8-week matrix for all 5 items with all six MRP rows.*

What changed, and why

The revised prompt makes 3 things explicit that the 1st prompt left implicit: it states the lead-time offset item by item instead of trusting the AI to read it correctly off the exhibit; it states the Week 1 scheduled-receipt rule as an instruction rather than background context; and it names lot-size rounding directly, rather than assuming the AI would treat Exhibit 1's lot sizes as a hard rule instead of a default guess. Once those 3 rules were spelled out, the AI's second output matched the manual solution exactly across all 5 items and all 6 MRP rows: same gross requirements, same scheduled receipts, same balances, same net requirements, same planned order receipts, and same planned order releases.

Part 2: Probe the Planning Policy

Prompt A - *Looking at Exhibit 1, what assumption did you make about the lot sizes (613, 1,226, 1,839, 2,452, 1,839) when you built the MRP? Did you treat them as fixed, non-negotiable batch sizes, or did you consider whether they were reasonable given the case's cost data?*

AI response: I treated the lot sizes as fixed inputs, exactly as listed in Exhibit 1, and rounded every order up to the nearest multiple. I did not question where those numbers came from or whether they still made sense. MRP mechanics take the lot-sizing rule as a given parameter, the same way it takes lead time or on-hand inventory as given, and the first prompt only asked me to run the matrix, not to evaluate the policy behind it.

Prompt B - *Are lot sizes really just input data on the same footing as on-hand inventory and lead time, or are they a planning policy that A-CAT could choose to change? Explain the difference*

AI response: There is a real difference. On-hand inventory and lead time are facts about the current state of the business; you observe them, you do not choose them. Lot size is different. It is a decision someone made, using some rule, possibly EOQ, possibly a round-number convention, possibly historical practice. Because it is a choice, it can be revisited whenever the cost structure, demand pattern, or supplier terms change. Exhibit 1 presents lot size as if it were static data, but it is really a stored policy decision, which is exactly why Menon is right to question it.

Prompt C - *Using the demand, setup cost, unit cost, and carrying cost in Exhibit 5, what additional lot-sizing analysis could I run to check whether these lot sizes are still reasonable?*

AI response: The direct check is the Economic Order Quantity formula, EOQ = square root of (2 x D x S / H), where D is average periodic demand, S is setup cost, and H is holding cost per unit per week (0.5% of unit cost, per Exhibit 5). Working average weekly requirement down through the bill of materials (P0110: 375/week; P0X and P0Y: 750/week each; P1X1: 3,000/week; P1X2: 2,250/week) gives the following EOQ figures against the case's given lot sizes:

Item	Avg. demand/week	Setup cost (S)	Holding cost/week (H)	EOQ	Case lot size
P0110	375	Rs 500	Rs 1.000	612	613
P0X	750	Rs 400	Rs 0.325	1,359	1,226
P0Y	750	Rs 350	Rs 0.175	1,732	1,839
P1X1	3,000	Rs 430	Rs 0.025	10,159	2,452
P1X2	2,250	Rs 400	Rs 0.075	4,899	1,839

P0110's current lot size is almost exactly its EOQ, so it is already close to cost-minimizing. P0Y's case lot sits a little above its EOQ. But P0X, P1X1, and P1X2 diverge more, especially P1X1, where the case lot is less than a quarter of the EOQ. That gap is worth investigating further, though, since EOQ assumes a steady demand rate and A-CAT's weekly demand is anything but steady.

Prompt D - *Generate a revised MRP scenario for P0X, P0Y, P1X1, and P1X2 using the EOQ lot sizes instead of the case's fixed lot sizes (keep P0110 unchanged, since its EOQ barely differs from 613). Then compare the two scenarios on ending inventory and total cost, setup cost plus holding cost, over the 8 weeks.*

Result -

Item: P0X OH: 500 LT: 1 week(s) Revised (EOQ) lot size: 1359								
Period	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Gross requirements	1226	1226		1226		1226		
Scheduled receipts	1359							
Projected available balance	633	766	766	899	899	1032	1032	1032
Net requirements	726	593		460		327		
Planned order receipts		1359		1359		1359		
Planned order releases	1359		1359		1359			

Item: P0Y OH: 340 LT: 1 week(s) Revised (EOQ) lot size: 1732								
Period	1	2	3	4	5	6	7	8
Gross requirements	1226	1226		1226		1226		
Scheduled receipts	1732							
Projected available balance	846	1352	1352	126	126	632	632	632
Net requirements	886	380				1100		
Planned order receipts		1732				1732		
Planned order releases	1732				1732			

Item: P1X1 OH: 500 LT: 1 week(s) Revised (EOQ) lot size: 10159								
Period	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Gross requirements	5436		5436		5436			
Scheduled receipts	10159							
Projected available balance	5223	5223	9946	9946	4510	4510	4510	4510
Net requirements	4936		213					
Planned order receipts			10159					
Planned order releases		10159						

Item: P1X2 OH: 600 LT: 1 week(s) Revised (EOQ) lot size: 4899								
Period	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Gross requirements	4077		4077		4077			
Scheduled receipts	4899							
Projected available balance	1422	1422	2244	2244	3066	3066	3066	3066
Net requirements	3477		2655		1833			
Planned order receipts			4899		4899			
Planned order releases		4899		4899				

AI response (cost comparison for 8 weeks):

Item	Case setup	Case holding	Case total	EOQ setup	EOQ holding	EOQ total	Cheaper
P0X	1,600	1,300	2,900	1,600	2,294	3,894	Case
P0Y	1,050	1,334	2,384	1,050	997	2,047	EOQ
P1X1	1,290	100	1,390	860	1,210	2,069	Case
P1X2	1,200	360	1,560	1,200	1,470	2,670	Case
Total			**8,234**			**10,681**	**Case**

Comparison and trade-offs

The EOQ-based scenario only wins for 1 item which is P0Y, trimming the batch slightly and saving about Rs 340 in total cost. For the other 3, EOQ does the opposite of what it is supposed to do: P0X and P1X2 end up carrying substantially more idle stock with no reduction in order count at all, so there is no offsetting saving. P1X1 does eliminate one order and saves Rs 430 in setup cost, but only by parking close to 10,000 units of stock at points in the schedule, nearly 20x its original peak inventory, for a net cost increase of roughly Rs 680. Added up, the case's own lot sizes cost about Rs 8,234 in setup and holding cost across the 8 weeks, versus Rs 10,681 under EOQ. In this specific window, the case's plan beats the textbook optimum.

That result would likely flip over a full year of steady average demand, which is the condition EOQ actually assumes. A-CAT's weekly demand for P0110 swings between 225 and 525

units, not a flat 375, and the case's lot sizes look tuned to that lumpiness rather than to a long-run average. Switching to the EOQ figures would only make sense if A-CAT expects demand to smooth out, if warehouse space and working capital can absorb much larger batches for P1X1 and P1X2, and if suppliers will actually sell in those exact quantities. Absent that, a plan built for this specific 8-week window beats a plan built for a hypothetical steady state, and Menon would be right to be skeptical of swapping in the EOQ numbers without checking feasibility first.

Final Reflection

This exercise makes a clean case for where AI genuinely helps in operations planning and where it does not. On calculation, AI is fast and reliable once rules are unambiguous: multiplying through a bill of materials, netting against on-hand inventory, and rounding to lot sizes are exactly the kind of repetitive arithmetic AI should be trusted to run at scale, far faster than doing it by hand across 5 items and 8 weeks. But the first attempt in Part 1 shows the limit of that trust: without a precise prompt, the AI quietly guessed at details the case had actually specified, such as lead time and how to treat Week 1, and produced a plausible-looking but wrong matrix. That is a decision-support gap, not a calculation one, and it only closed once the prompt spelled out the rules explicitly, which means a planner still has to know those rules well enough to catch the gap in the first place.

Part 2 pushes into managerial judgment. AI can compute an EOQ instantly and flag that the case lot sizes look economically odd for a couple of items. What it cannot do on its own is decide whether that oddity is a mistake or a sensible response to lumpy demand, warehouse limits, or supplier minimums, something the cost comparison here showed clearly favors the original plan over the textbook EOQ figures. Before Menon changes any MRP parameter, that judgment call, weighing textbook optimality against A-CAT's real operating constraints, has to stay with her, not the tool that ran the numbers.